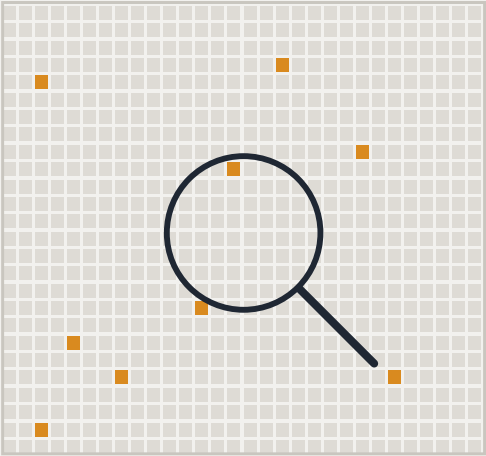


From 10K unlabeled SAE dimensions to the five that matter for one probe — ranked by verified counterfactual flips, then fixed

(a) Today: browse an SAE

manual · correlational · not actionable



10,240 dims — which ones does the hummingbird probe use?

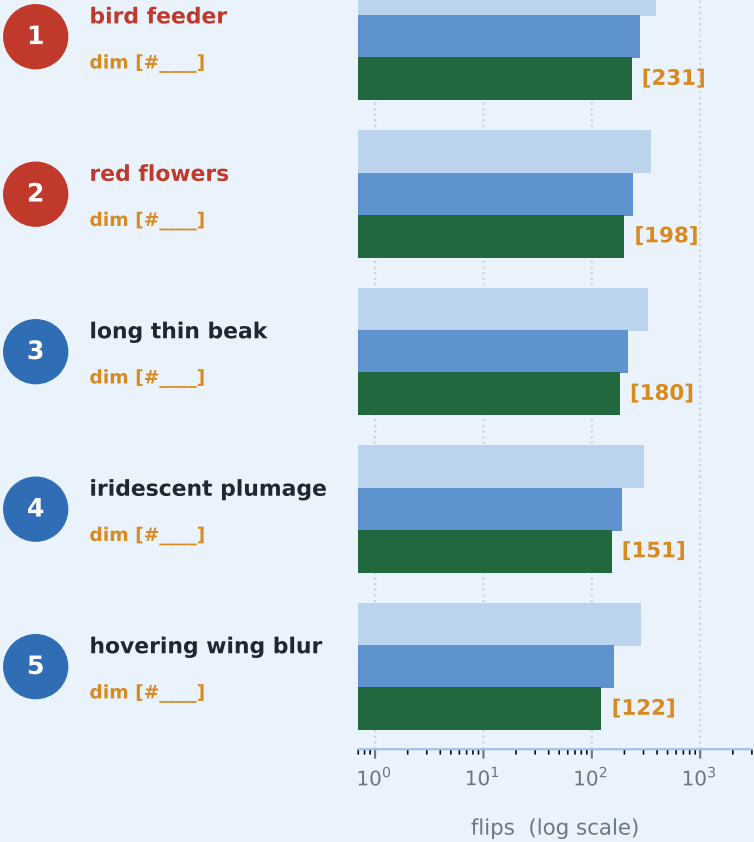
per dim: highest-activating images



✗ feeder or bird? unclear · no fix

(b) Rank all directions by verified flips

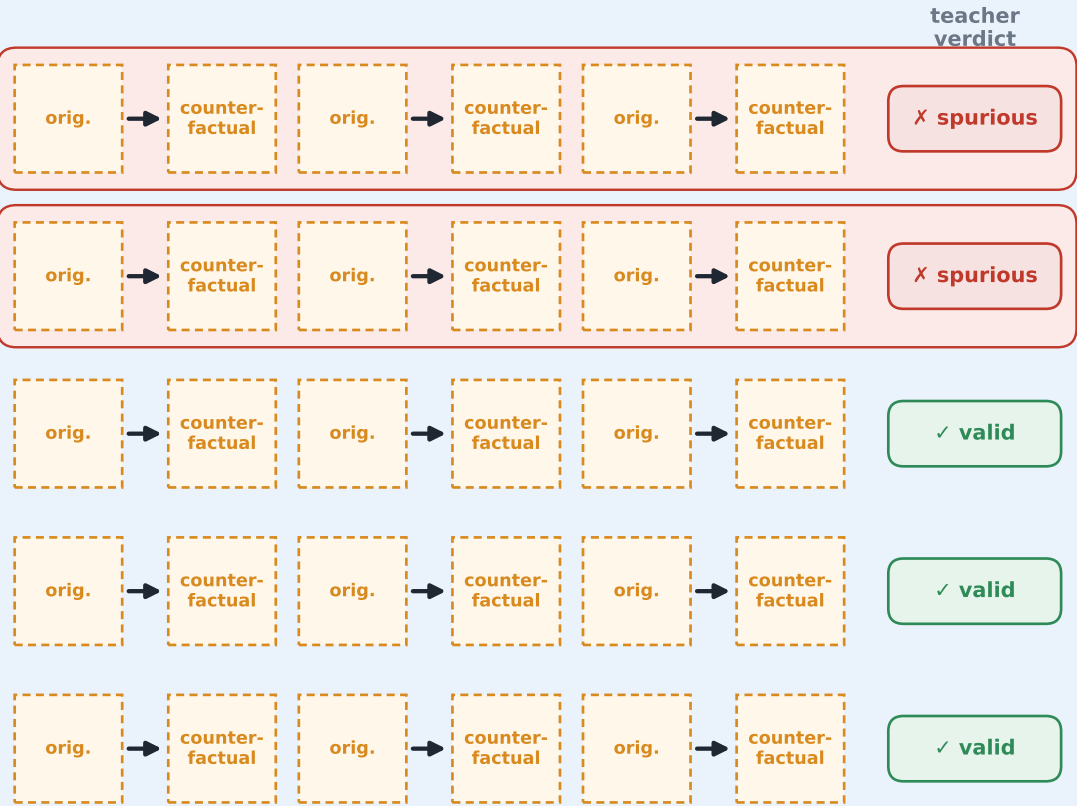
DINOv3 linear probe, ImageNet class “hummingbird”
public SAE, no SAE training · 512 images swept per direction



latent flips (distilled probe)
ambient flips (real classifier)
verified flips — rank key

(c) Counterfactuals of the top-5 only (3 verified flips per direction)

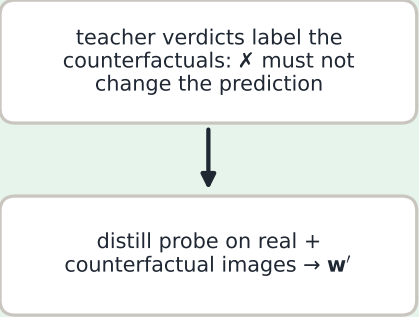
original $\mathbf{x}$ → counterfactual $\tilde{\mathbf{x}} = \text{decode}(\mathbf{z} + \delta \mathbf{v}_k)$ — the probe's decision flips on every pair



rank 1-2 reproduce the “bird feeder” / “red flowers” Clever Hans of Neuhaus et al. (2022) for hummingbird — found from the probe alone, with precise counterfactuals instead of highest-activating samples, and ranked among all directions in a single pass

(d) Fix the probe

counterfactual knowledge distillation (CFKD)



p(hummingbird) on feeder-only images

before [0.57]
after [0.13]

worst-group accuracy

before [61%]
after [79%]

✓ backbone untouched